

Lecture - 1 / Chapter 25 (Randall D. Knight)
Physic for Scientists and Eng.

Electric Charges and Forces

①

The electric force is one of the fundamental forces of nature. Lightning strike, electric forces can be wild and uncontrolled.

Interestingly, controlled electricity is the cornerstone of our modern, technological society.

Try to imagine to live without electricity!

Controlling forces is important to make it useful,

But how!

(i) We can control and manage this force?

(ii) What are the properties of electricity and electric forces?

(iii) How do we generate, transport and use electricity?

These are the questions we will explore throughout the semester in ~~our~~ lectures

We will start with the ELECTRIC

CHARGE.

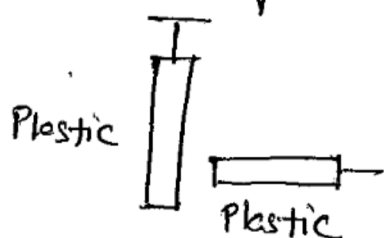
Developing a charge Model

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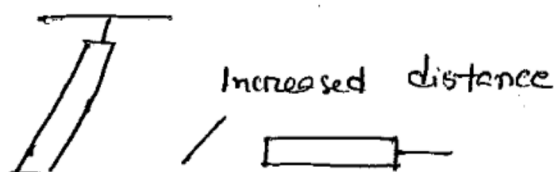
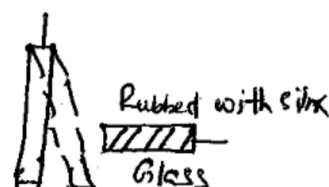
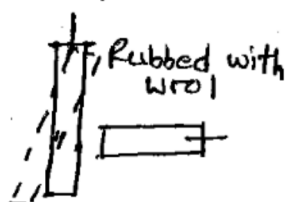
Our first goal is to develop a model for understanding electric phenomena in terms of charges and forces.

NOTE: The theory of electricity was well established long before the electron was discovered.

Experimenting with Charges



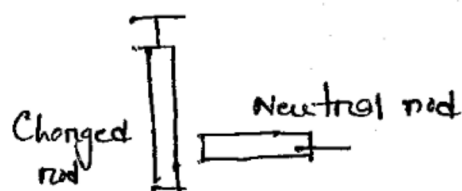
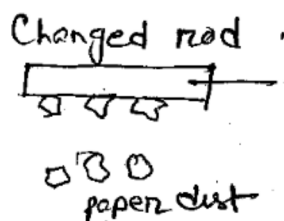
Set -1



○ The strength of these forces is greater for rods that have been rubbed more vigorously.

○ The strength of the forces decreases as the separation between the rods increases.

Set 2



(3)

□ First set of experiments found that charged objects exert forces on each other.

(1) The forces are sometimes attractive some time repulsive.

□ Second set of experiments found that there is an attractive force between a charged object and a neutral object.

⊛ How we can tell a object is neutral or charged.

1. Attractive force between a charged and neutral object, does not disclose that object is neutral.

2. But we can conclude this with the test

→ An important characteristics of any charged object appears to be that "a charged object picks up small pieces of paper." This behaviour provides a straight-forward test to answer the question — As: Is the object charged? — If object pass the test is charged; objects that fails the test neutral.

We can conclude from these experiments to (4) postulates:

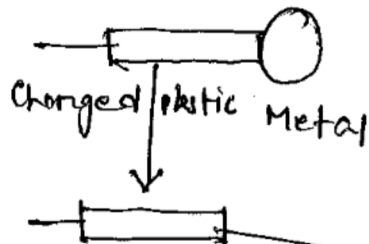
1. Frictional forces, such as rubbing, add something called charge to an object or remove it from the object.
2. There are only two kinds of charge - called as
a. Plastic Charge b. glass charge
3. Two like charges (plastic and/plastic on glass/and glass) and two opposite charges (plastic/and glass)
4. The force between two charges is a long-range force. The size of the force increases as the quantity of charge increases and decreases as the distance between the charges increases
5. Neutral objects have an equal mixture of both "plastic charge" and "glass charge".

If there is a third kind of charge different from plastic and glass - ~~what~~ how it will act?

Electric Properties of Materials

How different type of materials respond to charges?

(24)

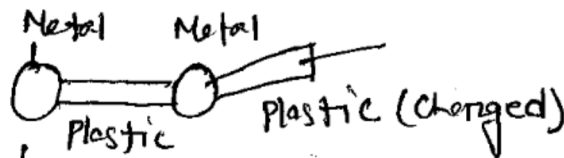


The metal sphere acquires 'Plastic Charge' and picks small pieces of paper.

Plastic rod no longer picks up small pieces of paper.

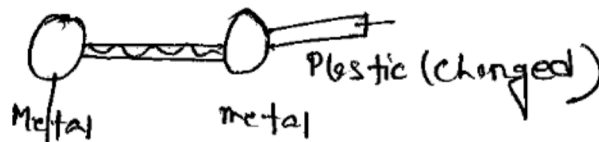
Charge can transfer

(25)



This metal ball remain neutral

(26)



This metal sphere acquires Plastic Charge.

Conclusions

1. Charge can transferred from one object to another, but only when the object touched. Removing charge is called discharging.

2. There are two types or classes of materials with very different electric properties. We call this → Conductors or insulators.

* Both insulators and conductors can be charged. What is the difference?

Charge

We have two charges (Plastic and glass) so far. But in modern science we rename them as positive charge and negative charge. Why?

Benjamin Franklin (one of the founding fathers of USA) was one of the first to make quantitative studies of electric phenomena rather than stay on qualitative observations.

He found that

① Charge behave like positive and negative numbers.

→ If a plastic rod is charged twice, by rubbing, and twice transfers charge to a metal sphere, the electric forces exerted by the sphere are doubled. That is $2+2=4$.

But →

The sphere is found to be neutral after receiving equal amount of 'plastic charge' and 'glass charge'.

That is $2-2=0$.

These experiments established an important property of charge. So what is negative and what is positive charge. It's entirely upto us. But by convention

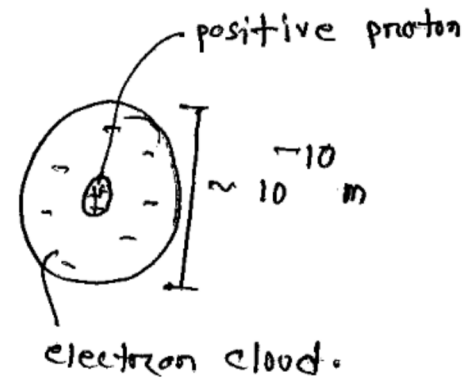
glass charge is positive; A long afterward, electron and proton were discovered and by convention electron represent negative charge.

Atoms and Electricity

(7)

The theory of electricity was developed without knowledge of atoms. For the contemporary perspective, we will discuss about the relevant characteristics of atoms and matter.

The figure shows that an atom consists of a very small and dense nucleus ($\sim 10^{-14} \text{ m}$) surrounded by much less massive orbiting electrons.



→ The electron's orbital frequencies are so enormous ($\sim 10^{15}$ revolution per second) that the electrons seem to form electron cloud of diameter $\sim 10^{-10} \text{ m}$, a factor of 10^4 larger than the nucleus.

A well-defined electron trajectory concept is no more exist and all we know are electron's size and shape of the electron cloud.

†!! wave-particle duality destroy the electron trajectory concept.

One of the most important discoveries is that charge is an inherent property of electron like mass. Electron is not possible an electron, without charge than it is to have an electron without mass.

— This atomic level unit of charge, called the fundamental unit of charge represented by the symbol 'e'. We will define the unit later.

The Micro/Macro Connection

NOTE-1: Electrons can move from one object to another when the objects are in contact, but neither electrons nor protons can leap through the air from one object to another. An object does not become charged simply from being close to a charged object.

NOTE-2: Neutral does not mean 'no charges' but instead, there is no net charge or net charge is zero.

A typical 1 cm^3 solid contains $\sim 10^{24}$ electrons and an equal number of protons. These are the huge number of particles and most of the solids are electrically neutral. A glass rod loses only $\sim 10^{10}$ electrons as it is charged by rubbing, this corresponds to only 1 electron out of 10^{14} .

Charge is represented by the symbol q . A macroscopic object such as a plastic rod has charge

$$q = N_p e - N_e e = (N_p - N_e) e \quad \left| \begin{array}{l} N_p \rightarrow \text{protons} \\ N_e \rightarrow \text{electrons} \end{array} \right.$$

$\Rightarrow 0$ means electrically neutral.

If $N_p > N_e \rightarrow$ positively charged

$N_e > N_p \rightarrow$ Negatively charged

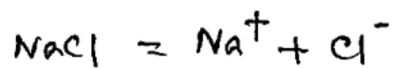
NOTE 3 The amount of charge on an object varies by small but discrete steps, not continuously (Multiple of e). This is called charge quantization.

⑦. In practice, objects acquire a positive charge by ^{not} gaining protons, which we expect, but by losing electrons. protons are extremely tightly bound within the nucleus and can't added or removed from atom.

On the otherhand, Electrons are loosely bound and can be removed without great difficulties. The process of removing an electron from electron cloud of an atom is called ionization. An atom that is missing an electron is called a positive ion. Its net charge is $q = +e$.

In otherhand, if some atoms can accommodate an extra electrons and thus become a negative ion, $q = -e$.

⑧ For example Table salt NaCl in water becomes



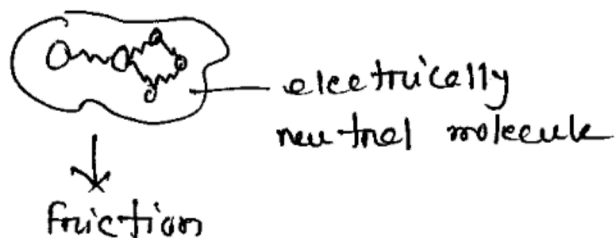
⑨ The force of friction (due to rubbing) cause molecular bonds at the surface to break as the two materials slides past each other.

Charge Conservation and Charge Diagrams.

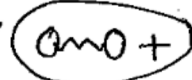
One of the important discoveries about charge is the law of conservation of charge. Charge is neither created nor destroyed. Charge can be transferred from one object to another as electrons and ions move about, but the total amount of charge remains constant.

For example! Charging a plastic rod by rubbing it with wool transfers electrons from the wool to the plastic as the molecular bonds break. The wool is left with a positive charge equal in magnitude but opposite sign in sign to the negative charge of the rod: $q_{\text{wool}} = -q_{\text{plastic}}$

Poodle model



Bonds were broken by friction.

 This half of the molecule lost an electron as the bond broke.
  Negative molecular ion
 This half of the molecule gained an extra electron as the bond broke.

Figure! Charging by friction usually creates molecular ions as bonds are broken.

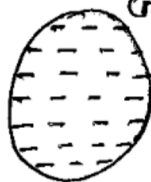
Charge Diagrams

Cross section of a conductor



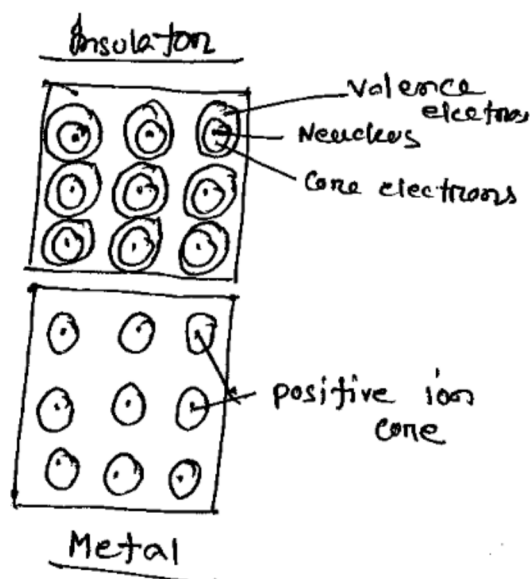
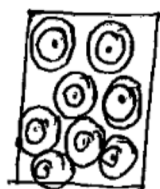
net positive charge
on surface

Cross section of an insulator



net negative charge in interior

Insulators and Conductors:



In metals, the outer atomic electrons are only weakly bound to the nuclei. As atoms come together to form a solid, these outer electrons become detached from their parent nuclei and are free to wander about through the entire solid. The solid as a whole remains electrically neutral, because we have not added or removed any electrons, but we can consider a sea of electrons with floating boats (ion cores).

Charging:

Insulators are often charged by rubbing.

The charge on the rod is immobile.

Metal usually can't be charged by rubbing, but from our previous experiment we show that a metal sphere can be charged by contact with a charged plastic rod.

→ The electrons in a conductor are free to move. Once charge is transferred to the metal, repulsive forces between the charges cause the electrons to move apart from each other. The electron sea takes an extremely short time to adjust in itself to the presence of the added charge, typically less than 10^{-9} sec.

→ For all practical purposes, a conductor responds instantaneously to the addition or removal of charge.

→ Other than this very brief interval during which the electron sea is adjusting, the charges in an isolated conductor are in static equilibrium. That is, the charges are at rest and there is no net force on any charge.

This condition is called Electrostatic equilibrium.

If, there were a net force on one of the charges, it would quickly move to an equilibrium point at which the force was zero.

Electro-Static equilibrium has an important consequences!

In an isolated conductor, any excess charge is located on the surface of the conductor. (*)

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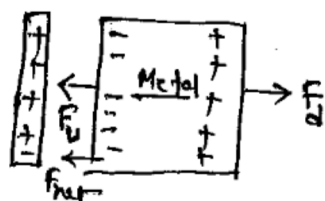
Discharging Any object that is physically connected to the earth through a conductor is said to be grounded.

→ To share charges to earth, eventually to prevent a voltage difference between earth and the object.

Charge Polarization:

So far we explain charging process, properties of insulator and conductor. How do charged objects of either sign exert an attractive force on a neutral object?

As electrons in conductor are free to move about, the whole electron sea shifts slightly towards an external positive charge. Although the metal as a whole is still electrically neutral, we say that the object has been polarized.



Charged polarization is a slight separation of positive and negative charge in a neutral object.

→ The net force due to charge separation is called a polarization force. (This not for the opposite charge in metal).